

Multiple Choice

1. **Answer:** C

Options: A) 2; B) 4; C) 8; D) 16

Note: Correct option: C - 8

2. **Answer:** A

Options: A) Supervised learning; B) Unsupervised learning; C) Clustering; D) None of the above

Note: Correct option: A - Supervised learning

3. **Answer:** A

Options: A) The polynomial degree; B) Whether we learn the weights by matrix inversion or gradient descent; C) The assumed variance of the Gaussian noise; D) The use of a constant-term unit input

Note: Correct option: A - The polynomial degree

4. **Answer:** A

Options: A) Lower variance; B) Higher variance; C) Same variance; D) None of the above

Note: Correct option: A - Lower variance

5. **Answer:** B

Options: A) The use of sampling with replacement as the sampling technique; B) The use of weak classifiers; C) The use of classification algorithms which are not prone to overfitting; D) The practice of validation performed on every classifier trained

Note: Correct option: B - The use of weak classifiers

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'It runs reasonably slow for multiple linear regression.'

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'PCA'.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.
10. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** Maximization. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.
12. **Answer:** either (a) or (b). Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key